PHYC10009 Foundations of Physics –Lecture 4

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SSLC Representative:

Equations of projectile motion - G 3.5

$$v_f = v_i + a\Delta t$$

$$\Delta x = v_i \Delta t + \frac{1}{2} a (\Delta t)^2$$

$$\Delta x = \frac{1}{2}(v_i + v_f)\Delta t$$

$$2a\Delta x = v_f^{\,2} - v_i^{\,2}$$

Horizontal motion (x direction):

$$a_x = 0$$

$$(v_x)_f = (v_x)_i$$

$$x_f = x_i + v_i \Delta t$$

Vertical motion (y direction):

$$a_y = -g$$

$$(v_y)_f = (v_y)_i - g\Delta t$$

$$y_f = y_i + (v_y)_i \Delta t - \frac{1}{2} g (\Delta t)^2$$

No need to remember these equations separately. Just remember the constant-acceleration equations!

Projectile motion problem solving - G 3.6

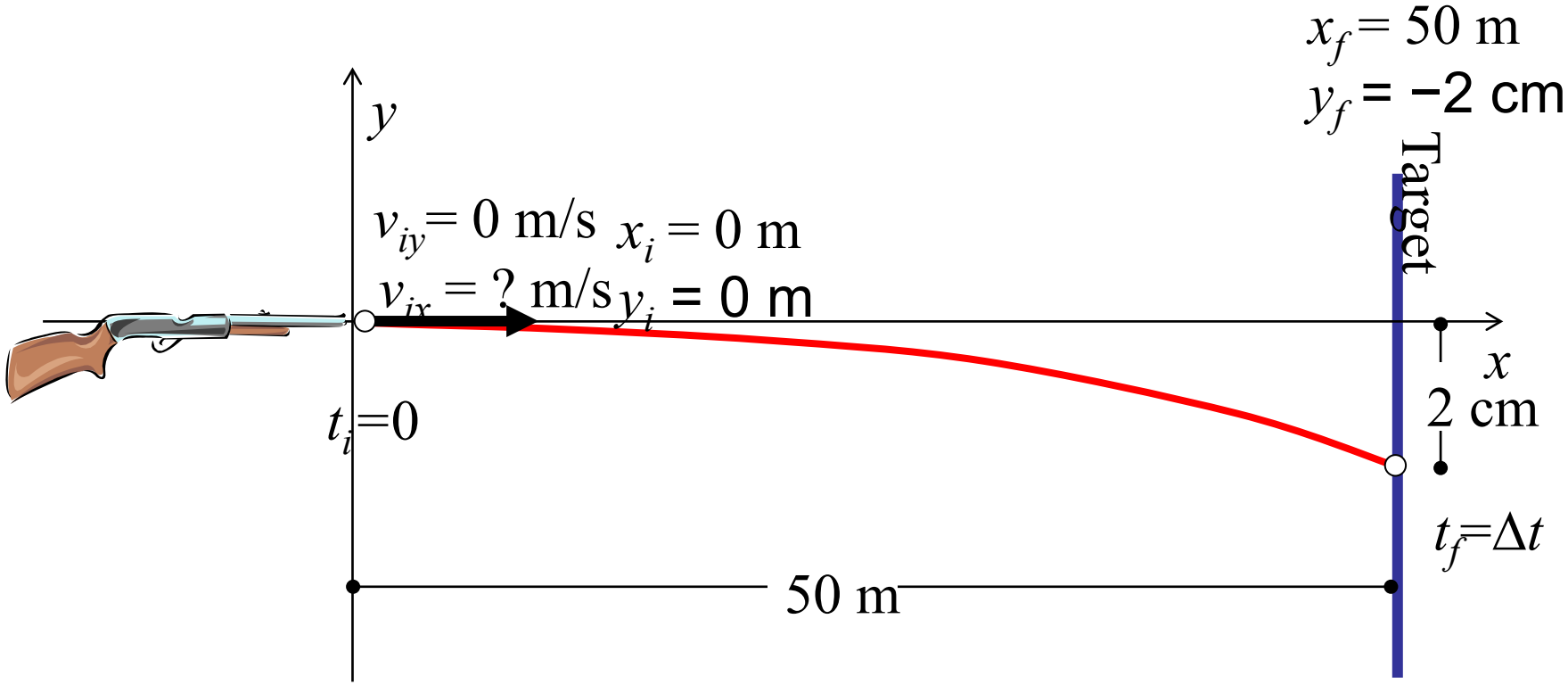
A rifle is aimed horizontally, i.e. with barrel parallel to the ground, at a target 50m away. The bullet hits the target 2cm below the aim point, i.e. below the height of the gun.

i. What was the bullet flight time?
ii. What was the bullet speed when it left the barrel of the rifle?

Assume bullet is point particle
Assume free-fall motion
No air-resistance

3.7 Projectile motion problem solving

Visualise:



Find: (i) Time elapsed Δt; (ii) v_{ix}

Projectile motion problem solving - G3.6

Solve: $y_f = \cancel{y_i}^{0} + \cancel{v_{iy}}^{0}\Delta t - \frac{1}{2}g(\Delta t)^2$

$$\Delta t^2 = \frac{-2y_f}{g} \Rightarrow \Delta t = \sqrt{\frac{-2\times(-0.02)}{9.8}} = 0.06 \text{ s}$$

$$x_f = \cancel{x_i}^{0} + v_{ix}\Delta t$$

$$v_{ix} = \frac{x_f}{\Delta t} = \frac{50}{0.06} = 800\ ms^{-1}$$

(i) $\Delta t = 0.06$ s

(ii) Speed $= 8 \times 10^2$ m/s

Projectile motion problem solving - G3.6

Assess:

Have you answered the question? ✓

Are units, signs and significant figures correct?

Units (i) time (s) and (ii) speed (m/s)

No sign issues

Data given with 1-2 significant figures ✓

Answer given with 1 sig fig

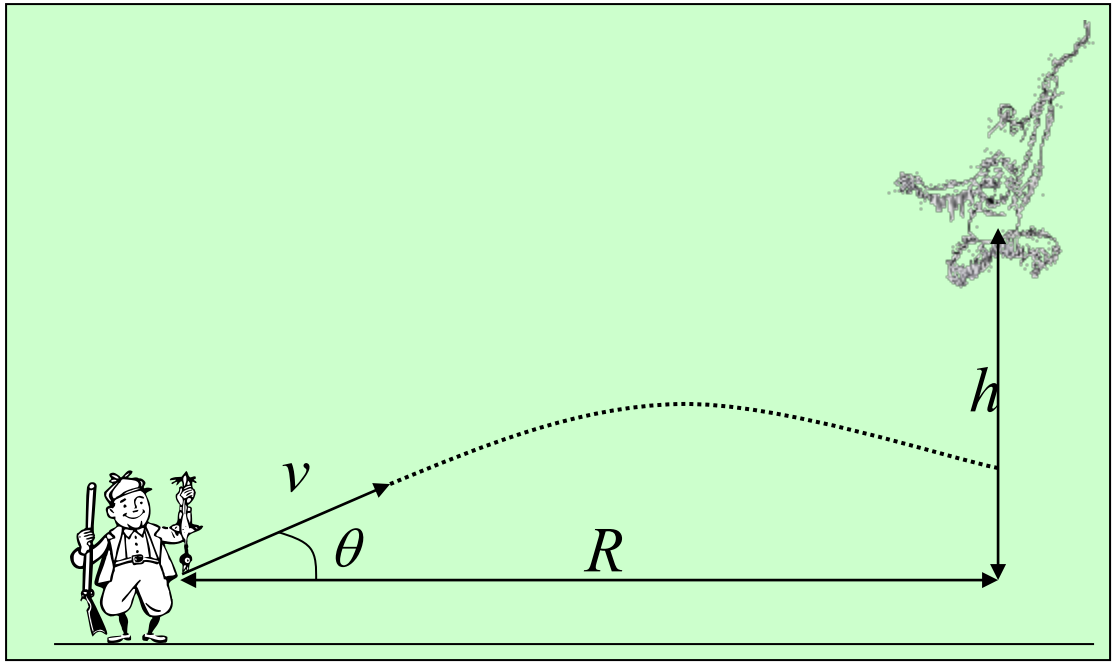
Vector answer where appropriate! ✓

Only speed and time requested, so no vector needed

Is answer reasonable? ✓

Yes. Would expect short time and fast speed since the bullet has dropped only 2cm in the time it takes to travel 50 m horizontally

Example: Monkey and dart - G3.6

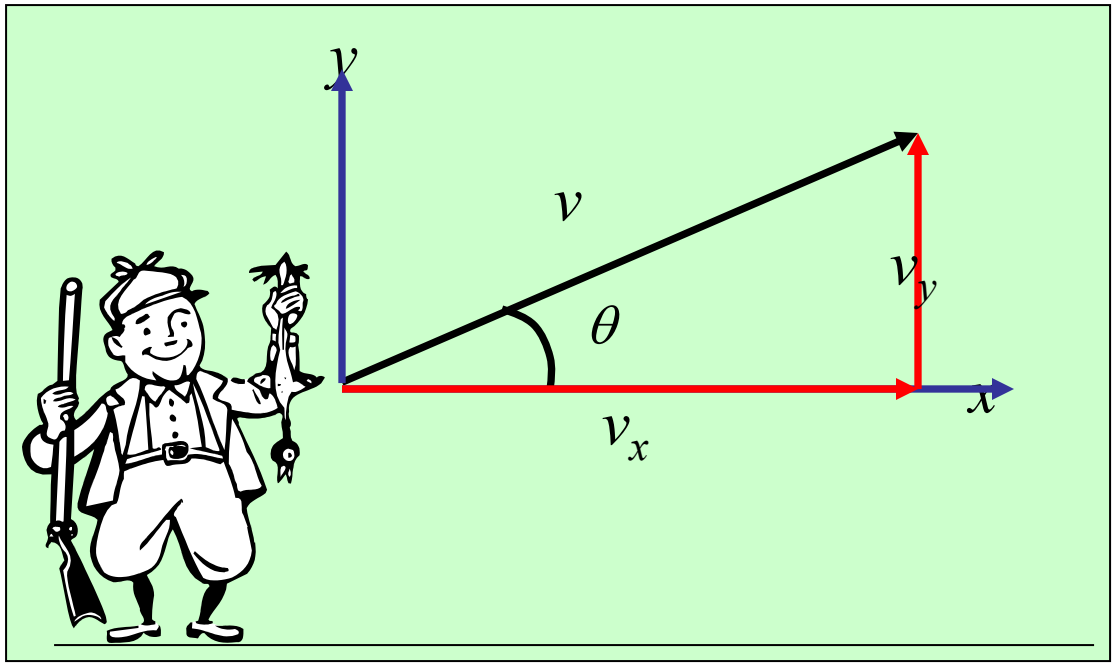


D

The friendly vet's gun has a fixed muzzle velocity v. At the same instant the vet fires the dart, the monkey drops from the vine. At what angle θ should the gun be fired to ensure that the dart hits the monkey?

Example: Monkey and dart - G3.6

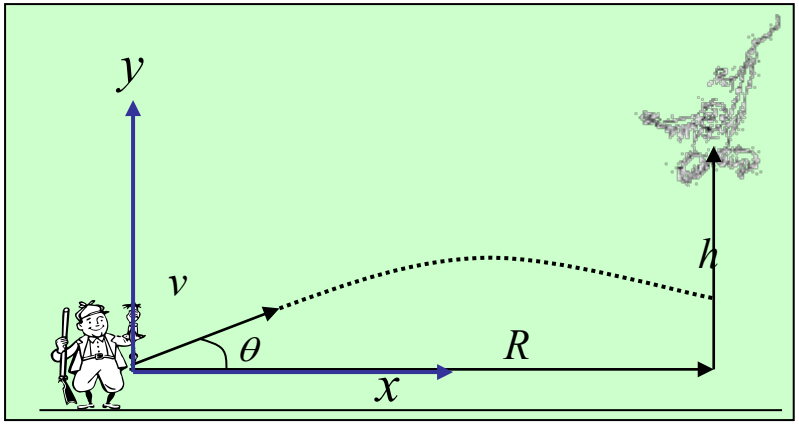
Components of initial velocity of dart



$$\cos\theta = \frac{v_x}{v}, \text{ so } v_x = v\cos\theta$$

$$\sin\theta = \frac{v_y}{v}, \text{ so } v_y = v\sin\theta$$

Example: Monkey and dart - G3.6



Horizontal motion

$$a_x = 0 \qquad x_f = x_i + v_i \Delta t$$

Dart: $x_f = R = (v\cos\theta)\Delta t$

Monkey: $x_f = x_i = R$

Vertical motion

$$a_y = -g \qquad y_f = y_i + (v_y)_i \Delta t - \frac{1}{2} g(\Delta t)^2$$

Dart: $y_f = 0 + (v\sin\theta)\Delta t - \frac{1}{2} g(\Delta t)^2$

Monkey: $y_f = h + (0)\Delta t - \frac{1}{2} g(\Delta t)^2$

$$R = (v\cos\theta)\Delta t$$

$$\Delta t = \frac{R}{v\cos\theta}$$

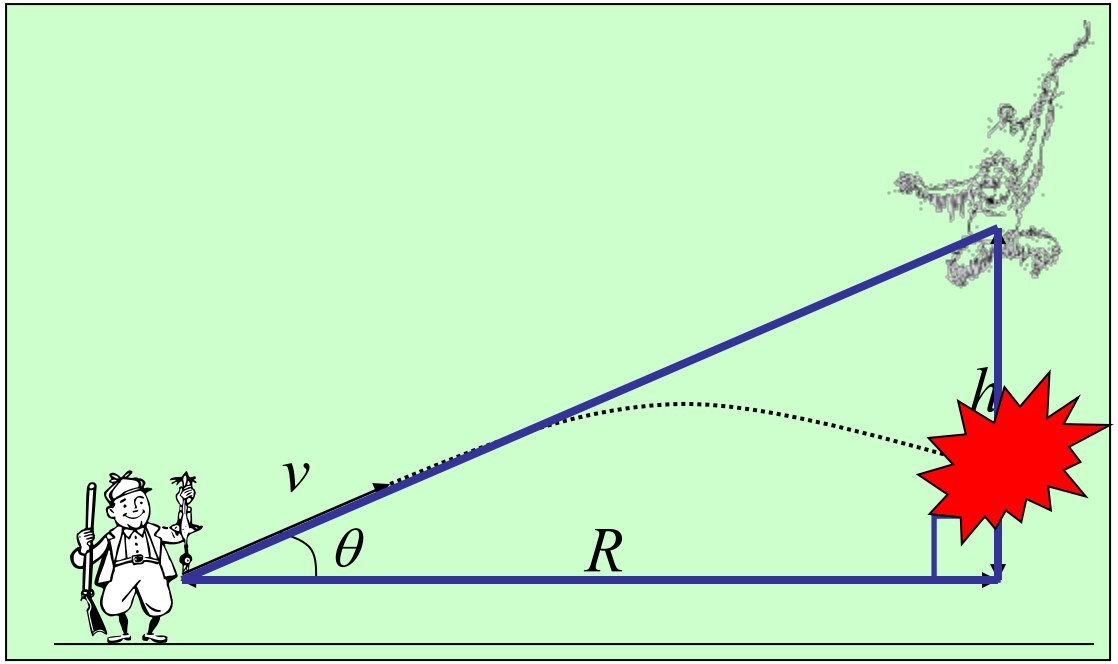
$$h = (v\sin\theta)\Delta t$$

$$h = v\sin\theta \frac{R}{v\cos\theta}$$

$$= R\tan\theta$$

$$\tan\theta = \frac{h}{R}$$

Example: Monkey and dart - G3.6



$$\tan\theta = \frac{h}{R}$$

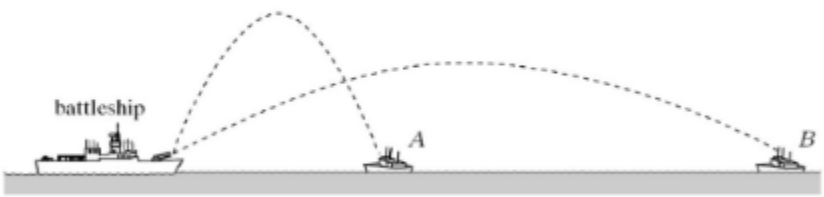
The angle we found is the same as the angle in right-angled triangle shown here.

Therefore, the vet should aim at the monkey's original position (before it drops).

Both the bullet and the monkey fall the same vertical distance in the same time interval.

Battleships

A battleship simultaneously fires two shells at enemy ships. If the shells follow the parabolic trajectories shown, which ship gets hit first?



A

Both at the same time

B

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Newton's laws of motion G-4

Newton I (law of inertia)

An object will continue to move with constant velocity (or, if at rest, remain at rest) unless a net force causes its velocity to change.

Newton II (acceleration is due to forces)

An object subject to a net force will accelerate in the direction of the net force at a rate proportional to the force and inversely proportional to the object's mass.

Newton III (action/reaction)

For every force on an object, there is a force of equal magnitude but opposite direction on some other object.

These are the inevitable consequence of a fundamental symmetry of Nature!

The Laws of Nature are the same in regions of space-time (2nd year Physics!)

What is a force? - G 4.1

A force

is a *push* or a *pull*.

can be a *contact* force or a *long range* force.

cf. one object pushes directly on another, force of gravity.

acts on an object and has an identifiable *cause* (e.g. a person or another object).

Isolated forces do not exist.

The cause of a force can be an inanimate object.

is a *vector*.

Force is applied to an object in a particular direction with a particular strength (magnitude).

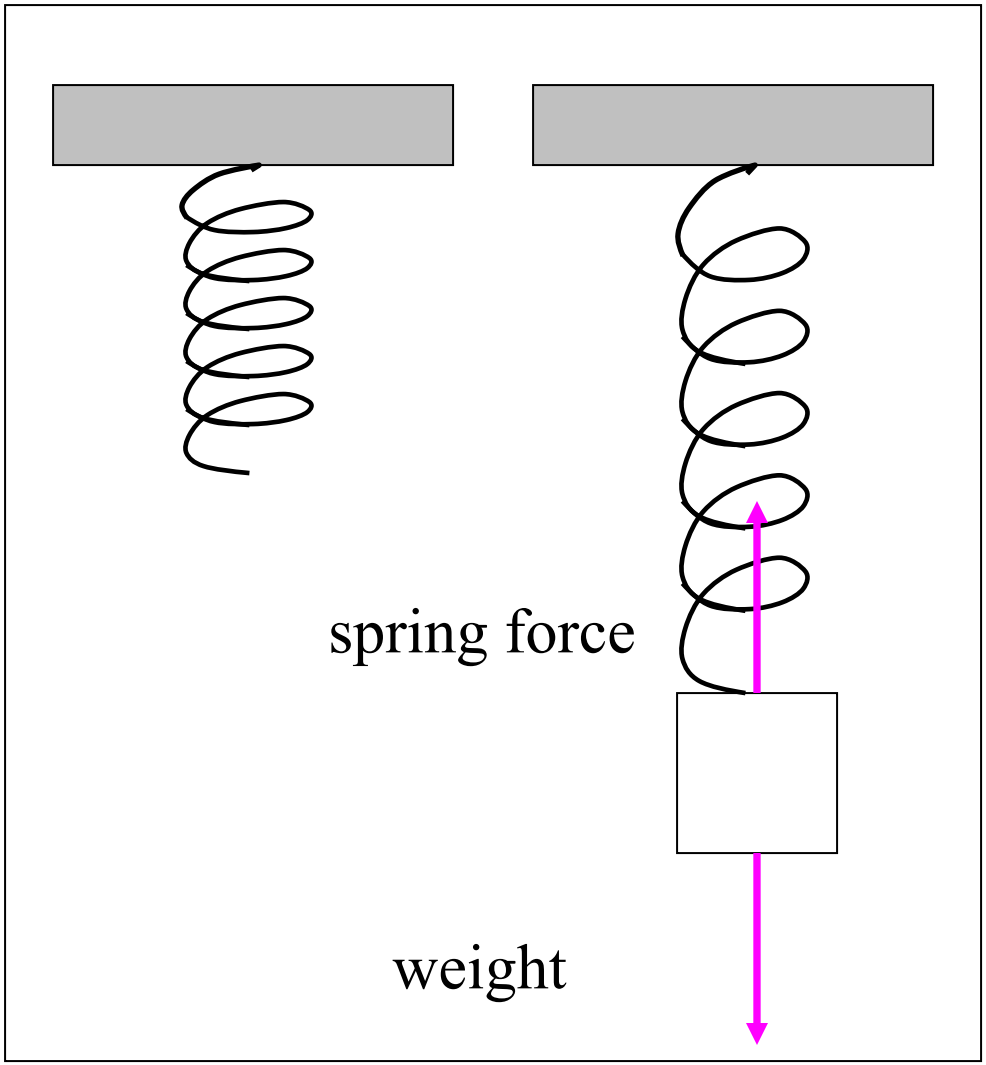
Examples of forces - G 4.1

A few examples of forces

- direct push or pull
- weight (gravity)
- spring force
- tension
- friction
- drag force (e.g. air resistance)
- electrostatic force
- magnetic force
- thrust (e.g. rocket)

We will look at many of these in more detail later…

Newton I – Zero net force - G4.2

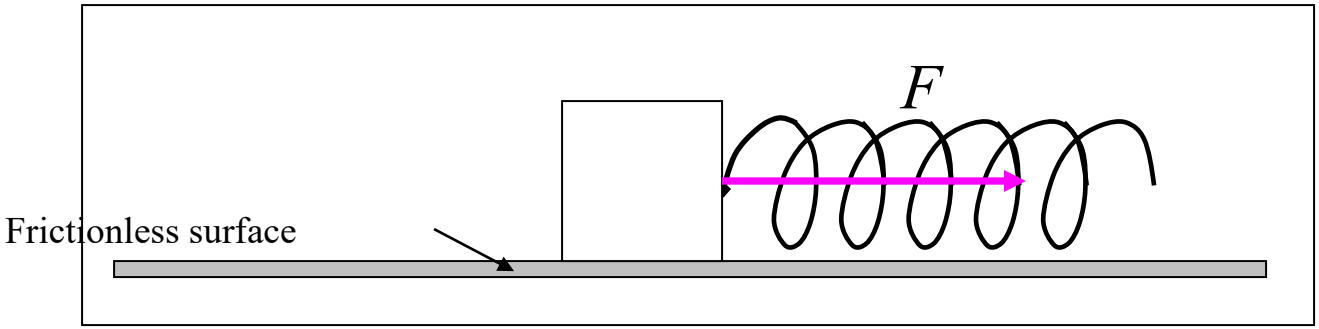


The net force is the vector sum of all the forces acting on an object.

Example:

- Block remains at rest on the stretched spring.
- Therefore, there is no net force (Newton I).
- The spring force and the weight force add as vectors to give zero.

Newton II – Net force and acceleration - G4.3,4.4



Experiment:

Apply different forces *F* to an object.

Can measure *F* by measuring the extension of the spring.

Observe: Applying force causes the object to *accelerate*.

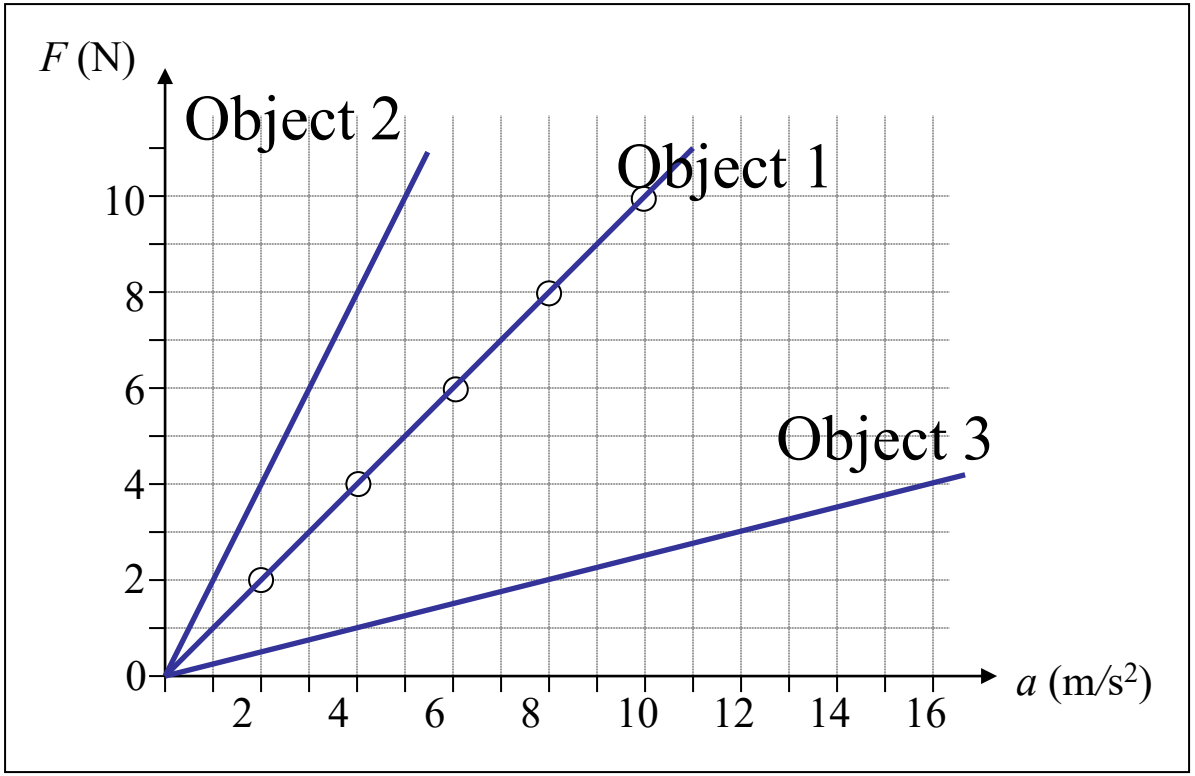
Measure the acceleration *a* for various forces *F*.

To measure acceleration, film the motion and calculate acceleration from the motion diagram, or use a motion sensor (as in first-year labs).

Try other objects, too.

D Newton's cart

Newton II – Net force and acceleration - G 4.4



Result:

Acceleration is directly proportional to the applied force, but the graph (sometimes) has different gradients for different objects.

$$a \propto F$$

or

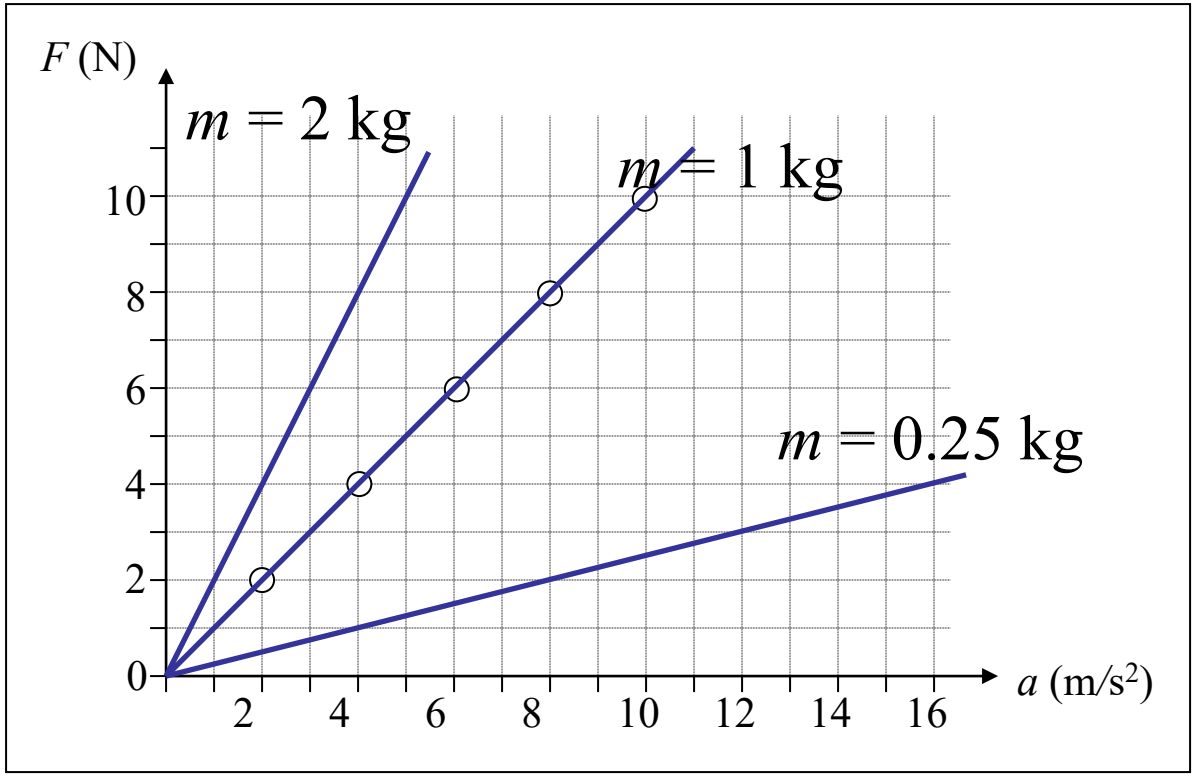
$$F = (\text{constant}) \times a$$

Newton II – Net force and acceleration - G 4.4

$$F = (\text{constant}) \times a$$

The constant is the gradient of the graph.

We *define* the **mass** of an object to be gradient of the force vs acceleration graph.



So $\vec{F} = m\vec{a}$ or

$$\vec{a} = \frac{\vec{F}}{m}$$ Newton's second law of motion

Newton II - G 4.4

effect

cause

$$\vec{a} = \frac{\vec{F}_{net}}{m}$$

Newton's second law of motion

SI unit of force:

kg

m s^{-2}

Newton (N)

$$\vec{F} = m\,\vec{a}$$

$$1\ \mathrm{N} \equiv 1\ \frac{\mathrm{kg\ m}}{\mathrm{s}^2}$$

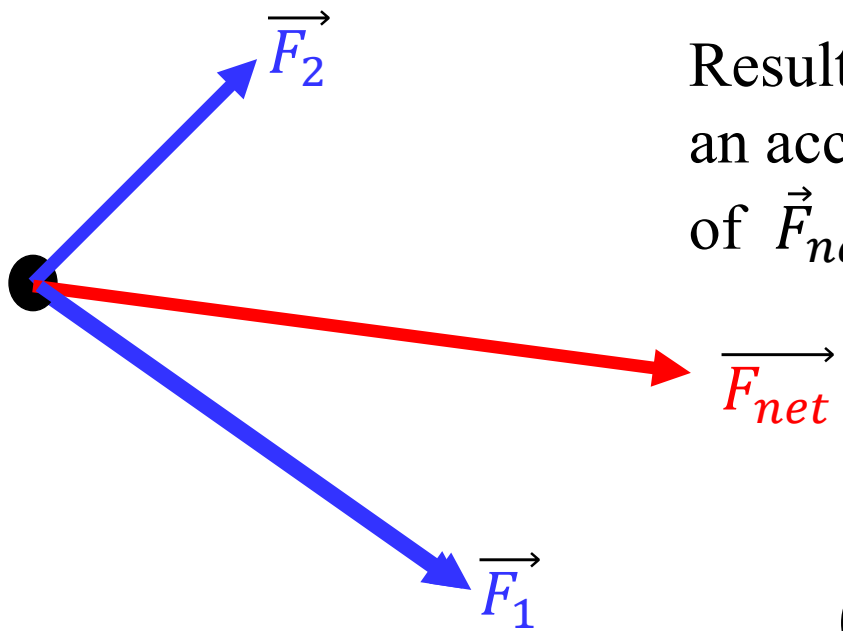
Force is a Vector!

effect

cause

$$\vec{a} = \frac{\vec{F}}{m}$$

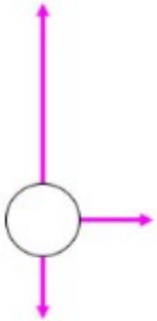
Newton's second law of motion



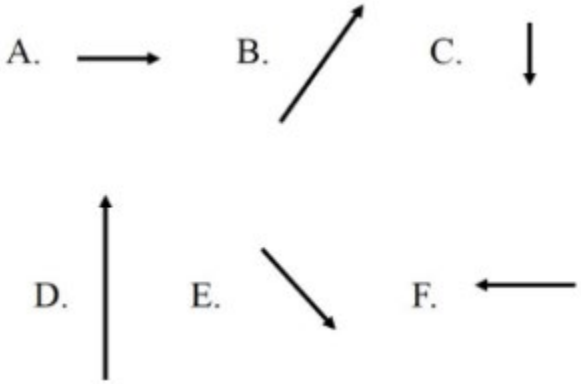
Result of $\vec{F}_1 and\ \vec{F}_2$ is an acceleration in direction of $\vec{F}_{net}$

D Vector addition of forces

Forces as Vectors



Three forces act on an object. In which direction does the object accelerate?



A

B

C

D

E

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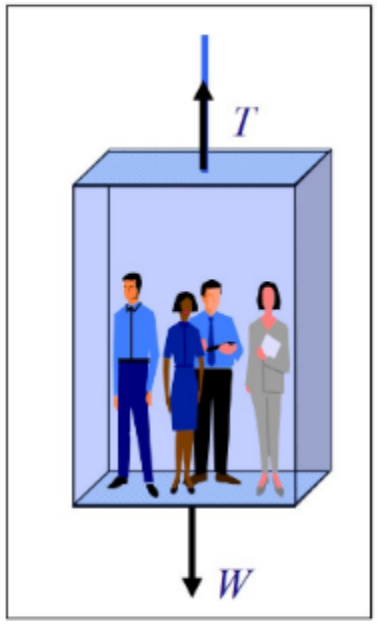
Elevators

An elevator is moving upwards at a constant speed. Which of the forces acting on the elevator has the greatest magnitude?



The tension force T from the Cable

The weight force W

Both forces have equal magnitude

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Summary of Lecture 4

The net force is the vector sum of all the forces acting on an object.

The net force causes a body to accelerate with magnitude inversely proportional to its mass

effect

cause

$$\vec{a} = \frac{\vec{F}_{net}}{m}$$

Newton's second law of motion

All forces exist as action/reaction pairs. There are no isolated forces in Nature.